# 7 Primes and Factorization

## 7.1 Primes and Factors

Over two millennia ago already, people all over the world were considering the properties of numbers. One of the simplest concepts is prime numbers. We use the notation $\mathbb{N}$ to denote the nonnegative integers. That is, $\mathbb{N} = \{0, 1, 2, \ldots\}$. Unless otherwise specified, we restrict our attention here to the set $\mathbb{N}$ throughout this chapter.

- We say $a$ is a **factor** or **divisor** of $b$, and $b$ is a **multiple** of $a$, if $a$ goes into $b$ without a remainder. For example, the factors of 6 are 1, 2, 3 and 6.
- A number greater than 1 with only itself and 1 as factors is called a **prime** number. Otherwise it is **composite**. (For the record, 1 is neither prime nor composite but is a **unit**.) For example, 19 is prime, but 21 is composite.
- The **gcd** (greatest common divisor) of two numbers, also known as the highest common factor, is the largest number that is a factor of both numbers. For example, $\gcd(6, 14) = 2$.
- Two numbers are **relatively prime** if they have no common factor apart from 1. That is, their gcd is 1.

If a number is composite, then it is a product of two nontrivial factors (nontrivial here meaning neither is 1). And these factors, if composite, are themselves products of factors. Eventually we reach primes. Thus any integer has a **prime factorization** (and the factorization is unique). For example, the prime factorization of 60 is $2^2 \times 3 \times 5$. This fact is sometimes called the "Fundamental Theorem of Arithmetic":

**Theorem 7.1** *Every integer $a > 1$ can be expressed uniquely as a product of primes.*

While this theorem may sound obvious, if one is being careful, this fact actually needs several lines to prove it. The reason for the name of the theorem is that there are many places in mathematics where a form of multiplication is defined, and one of the first questions asked is about uniqueness of factoring.

The prime factorization of a number can be used to determine the number of factors it has.

EXAMPLE 7.1. *How many factors does* 9000 *have?*

Its prime factorization is $3^2 \times 2^3 \times 5^3$. Since we can specify a divisor by independently specifying the power of 3, the power of 2 and the power of 5, this means that 9000 has $3 \times 4 \times 4$ factors.

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## 7.2 Euclid's Algorithm

If we have the prime factorization of the two numbers, then there is a straightforward recipe to calculate their gcd:

> *take the smaller power of each common prime and multiply together.*

For example, consider $\gcd(36, 120)$. Since $36 = 2^2 \times 3^2$ and $120 = 2^3 \times 3 \times 5$, their gcd is $2^2 \times 3 = 12$.

Surprisingly, knowing the factorizations is not necessary, as was known to Euclid and other ancients. To explain this, we need to introduce a new operator called "mod".

Let $d$ be a positive integer. For $c \in \mathbb{N}$, the value $c \bmod d$ is the remainder when $c$ is divided by $d$. For example,

> $c \bmod d = 0$ if and only if $d$ is a multiple of $c$.

Here is **Euclid's Algorithm** for the greatest common divisor of $a$ and $b$:

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GCD(a,b)
    if b is a factor of a, then return b
    else return gcd(b, a mod b)
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This is a recursive algorithm. The procedure is guaranteed to terminate, since the $b$-value decreases each time.

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EXAMPLE 7.2. *What is the* gcd *of* 33 *and* 12*?*

$\gcd(33, 12) = \gcd(12, 9) = \gcd(9, 3) = 3$.

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But we need to show that the procedure always gives the correct value.

If $b$ is a factor of $a$, then clearly their gcd is $b$. So we need to show that when $b$ is not a factor of $a$, that

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Well suppose $a = qb + r$ where $r = a \bmod b$. It follows that, if $c$ is a factor of both $b$ and $r$, then $c$ is a factor of $qb + r$ as well, and so it is a factor of $a$. Conversely, we can re-arrange the equation to $r = a - qb$, and so if $c$ is a factor of both $a$ and $b$, then it is a factor of $r$ as well. This shows that the set of common factors of $a$ and $b$ is the same as the set of common factors of $a$ and $r$. In particular, the greatest element in the two sets must be the same.

Euclid's algorithm can be extended to prove the following result, and indeed to construct the $s$ and $t$ the theorem claims exist:

**Theorem 7.2** *If $c$ is the* gcd *of $a$ and $b$, then there exist* integers *$s$ and $t$ such that*

$$s \times a + t \times b = c.$$

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EXAMPLE 7.3.

For 12 and 33, we have $3 \times 12 + (-1) \times 33 = 3$.

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We omit the proof of the theorem.

► **For you to do!** ◄
1. *Use factorization to calculate the gcd of* 100 *and* 240.
2. *Use Euclid's algorithm to calculate the gcd of* 100 *and* 240.

## 7.3 Unsolved Problems

While we have come very far in number theory in thousands of years of effort, there are still many unsolved problems, some of which are easy to state. For example:

**Conjecture 7.3** *(Goldbach's conjecture) Every even number at least* 4 *is the sum of two primes.*

**Conjecture 7.4** *(Twin prime conjecture) There are infinitely many pairs of prime that differ only by* 2 *(such as* 11 *and* 13*).*

There is a lot of "evidence" to support Goldbach's conjecture. Indeed, it appears that as the even number get larger, it is the sum of two primes in many ways. But a proof remains elusive.

Another question is how to find the factors of a number. Is there an algorithm which runs in time proportional to some polynomial of the number of digits of a number and is guaranteed to find the factorization? Only recently did mankind find such an algorithm which tests whether a number is prime or composite (but the "proof" of compositeness does not find the factorization. . . ).

***Exercises***

7.1. List all factors of 945.

7.2. How may factors are there of:

(a) 21

(b) $245,000$

(c) $2^{15}$

7.3. Define $D(n)$ as the number of factors of $n$. For example, the primes are the $n$ such that $D(n) = 2$.

(a) Determine all $n$ such that $D(n) = 3$.

(b) Determine all $n$ such that $D(n) = 4$.

7.4. A famous mathematician once noticed that the formula $f(n) = n^2 - n + 41$ yields primes for small values of $n$. For example, when $n = 1$ he calculated $f(1) = 41$, which is prime. When $n = 2$ he calculated $f(2) = 43$ and this is prime. Test the formula for $n = 3$, 4,and 5. Are the results prime? Does the formula always yield primes? Prove your answer.

7.5. Calculate the gcd of:

(a) 91 and 287.

(b) $12^{100}$ and $100^{12}$.

7.6. Let $a$ be a positive integer. Prove that $2a + 1$ and $4a^2 + 1$ are relatively prime.

7.7. Determine $\gcd(10^{10}, 20^{20})$.

7.8. Prove that if $bc$ is a multiple of $a$ and $\gcd(a, b) = 1$, then $c$ is a multiple of $a$.

7.9. Show that the gcd operation is associative. That is: $\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c)$.

7.10. (a) Show that $a^k - 1$ is a multiple $a - 1$ for any positive $a$ and $k$.

(b) Show that if positive $m$ and $n$ are not relatively prime, then $2^m - 1$ and $2^n - 1$ are not relatively prime.

(c) Show that if positive $m$ and $n$ are relatively prime, then $2^m - 1$ and $2^n - 1$ are relatively prime.

7.11. (a) Show that if $n = pq$ for distinct primes $p$ and $q$, then the number of integers from 1 up to $n$ that are relatively prime to $n$ is $(p-1)(q-1)$.

(b) Show that if $n = p^2$ for prime $p$, then the number of integers from 1 up to $n$ that are relatively prime to $n$ is $p(p-1)$.

(c) Generalize this. Let $\phi(n)$ be the number of values between 1 and $n$ that are relatively prime to $n$. Develop a formula for $\phi(n)$ based on the prime factorization of $n$.

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***Solutions to Practice Exercises***

1. $100 = 2^2 \times 5^2$ and $240 = 2^3 \times 3 \times 5$. So gcd $= 2^2 \times 5 = 20$.

2. $(240, 100) \rightarrow (100, 40) \rightarrow (40, 20) \rightarrow 20$